Threshold autoregression: SETAR models and limit cycles

Why do the lynx rise for six years and crash in three? Tong and Lim (1980), reproduced.

Tested against	Tong and Lim (1980), *Journal of the Royal Statistical Society: Series B*, eqs. 9.1–9.3 and Figs. 8, 10 and 12 (pp. 259–262): the Canadian lynx trappings of 1821–1934 (PollisDatasets `lynxmackenzie`) and Wolf's yearly sunspot numbers of 1700–1955 (R's `sunspot.year`, read into memory and not saved).
Missing in Julia	the self-exciting threshold autoregressive model (SETAR), its identification by AIC, and its eventual forecasting function and limit cycles. No registered Julia package fits it.
Notebook	**Compose** › **New Methods** › **Browse Methods**: *Threshold autoregression: SETAR models and limit cycles* (`ThresholdAR.ipynb`)
Result	68 of 70 printed values to every digit, the other two in the last. Every coefficient of the three models to 4 decimals.

The problem

Many cycles in nature and in economies are not symmetric: the Canadian lynx population climbs for about six years and collapses in three; sunspot activity rises faster than it falls. A linear autoregression cannot produce this. Its cycles are symmetric, and without noise it dies out to its mean. Tong and Lim (1980) let the autoregression change with the level of the series itself: one linear model below a threshold, another above it. So simple a nonlinearity gives asymmetric cycles, limit cycles and jumps.

The method

In SETAR$(2; k_1, k_2)$ with delay d and threshold r, the series follows

$$x_t = a_0^{(j)} + a_1^{(j)} x_{t-1} + \cdots + a_{k_j}^{(j)} x_{t-k_j} + \varepsilon_t^{(j)}, \qquad j = 1 \text{ if } x_{t-d} \le r,\ j = 2 \text{ otherwise},$$

and each regime is fitted by least squares to its own observations (`setar`). `identify` follows the paper's Section 8: the candidate thresholds are the 30–70 per cent points of x_{t-d}, each regime's order is the one with the smallest AIC, $n_j \ln(\mathrm{RSS}_j/n_j) + 2(k_j + 1)$, and the threshold the one with the smallest total. `selectdelay` compares delays by the normalised AIC, the AIC per observation fitted. `eventual` continues the series by the model without noise, the paper's eventual forecasting function; `limitcycle` finds the cycle it settles into and `risedescent` the rise and descent of each local cycle. `predict` gives one-step-ahead predictions.

What the tests show

Test	What is compared	Agreement
Eq. 9.1	lynx SETAR(2; 8, 3): 13 coefficients	every one to 4 decimals
Eq. 9.2	sunspots 1700–1920, SETAR(2; 4, 12): 18 coefficients	every one to 4 decimals
Eq. 9.3	sunspots 1837–1924, SETAR(2; 4, 2): 8 coefficients	every one to 4 decimals
p. 259	lynx limit cycle, period 9: 9 values	8 to every digit, 1 in the last
p. 261	sunspot cycle: period 31, rise and descent of 3 local cycles	all 7
Eqs. 9.1–9.3	variances and NAICs: 12 values	11 to every digit, 1 in the last
Figs. 8–12	lynx root mean RSS, two sunspot prediction RMSEs	2 to every digit, 1 in the last

The last-digit differences, side by side (Julia against the paper):

- The lynx limit cycle: 3.22805 vs 3.2281.
- The NAIC of eq. 9.3: 5.08178 vs 5.0817, which looks truncated rather than rounded.
- The one-step-ahead RMSE of eq. 9.3 over 1925–1944: 9.23414 vs 9.2342.

Two things the paper does not say

The variances and NAICs are those of a smaller model

The residual variances printed under each equation, and the NAICs of the figures, are not those of the models as stated. They are those of the same models with one lag fewer in each regime, and the NAIC penalises them as if they had the

orders stated. All three models show it, to every printed digit: eq. 9.2's variances 254.64, 66.80 and 153.71 are those of SETAR(2; 3, 11), and SETAR(2; 4, 12) gives 252.64, 66.55 and 152.65. The coefficients, the root mean RSS and the prediction errors are those of the models as stated: the 1921–1955 RMSE is 12.17397 for SETAR(2; 4, 12), as printed, and 12.35812 for SETAR(2; 3, 11). It looks like an off-by-one in the program that printed the summaries.

The lynx cycle is printed backwards

Read forwards, the printed lynx cycle rises for three years and falls for six; the text, and the model, rise for six and fall for three. The paper lists the cycle from its smallest value backwards.

Still open

The identification does not arrive at the published models by itself: for the lynx the smallest NAIC is at $d = 3$, SETAR(2; 5, 9), and at $d = 2$ the search prefers SETAR(2; 10, 2). Tong and Lim chose with the eventual forecasting function, the prediction errors of held-out observations and judgement as well (pp. 258–259); `selectdelay` returns every candidate beside the one with the smallest NAIC. The published threshold itself, 3.1163, is on the grid: it is the 60 per cent point of x_{t-2}, the 1902 count of 1,307 lynx.

Further considerations

- **More regimes and other thresholds.** The paper's TARSO and TARSC models, with an outside series or several series in the threshold, need data the paper does not give in full; they would need a published table with its data before they become a new method here.
- **Inference on the threshold.** Testing linearity against SETAR, and confidence sets for r, came later (Chan, Tsay, Hansen); each would need its own published table to be tested against.
- **Upstream.** SETAR would sit naturally in a Julia time-series package; the three equations are a ready-made test for such a contribution.

References

Tong, H. and Lim, K. S. (1980). Threshold autoregression, limit cycles and cyclical data. *Journal of the Royal Statistical Society: Series B* 42(3), 245–268. doi:10.1111/j.2517-6161.1980.tb01126.x

Elton, C. and Nicholson, M. (1942). The ten-year cycle in numbers of the lynx in Canada. *Journal of Animal Ecology* 11(2), 215–244.

Campbell, M. J. and Walker, A. M. (1977). A survey of statistical work on the Mackenzie River series of annual Canadian lynx trappings for the years 1821–1934 and a new analysis. *Journal of the Royal Statistical Society: Series A* 140(4), 411–431.

Data: the lynx counts from PollisDatasets (`lynxmackenzie`); the sunspot numbers as distributed with R (`sunspot.year`), read from R's source and not saved, since SILSO's licence does not allow them to be distributed with Pollis.

Methods written and tested with Claude Code (Anthropic) in Pollis, from the published papers and their data.
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